

Unit 3.2 Product and Quotient rule p183-188

Note Title

2016/04/10

Find:

$$\begin{aligned} 1. \quad \frac{d}{dx} x(\sqrt{x} + e) &= \frac{d}{dx} (x\sqrt{x} + ex) \\ &= \frac{d}{dx} (x^{3/2} + ex) \\ &= \frac{3}{2}x^{1/2} + e \\ 2. \quad \frac{d}{dx} x^2 e^x &= ? \\ &\quad \text{product.} \end{aligned}$$

Product rule p184 Know it!!!

If f and g are both differentiable, then
$$\frac{d}{dx} [f(x)g(x)] = \underline{f(x)} \frac{d}{dx} g(x) + \underline{g(x)} \frac{d}{dx} f(x).$$

Proof: Different from textbook.

$$\begin{aligned} \frac{d}{dx} [f(x)g(x)] &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\underbrace{f(x+h)g(x+h) - f(x)g(x)}_{\text{blue}} + \underbrace{f(x+h)g(x) - f(x+h)g(x)}_{\text{green}}}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)[g(x+h) - g(x)] + g(x)[f(x+h) - f(x)]}{h} \\ &= \lim_{h \rightarrow 0} f(x+h) \left[\frac{g(x+h) - g(x)}{h} \right] + \lim_{h \rightarrow 0} g(x) \left[\frac{f(x+h) - f(x)}{h} \right] \\ &= f(x) \frac{d}{dx} g(x) + g(x) \frac{d}{dx} f(x) \quad \square \end{aligned}$$

In prime notation

$$(fg)' = f'g + fg'$$

Now try

$$\frac{d}{dx} x^2 \cdot e^x = \underline{\underline{2x}} e^x + x^2 \underline{\underline{e^x}}$$

Examples: Use the product rule to find y' .

$$\textcircled{1} \quad y = x^2(x^3 + e^x)$$

$$y' = 2x(x^3 + e^x) + x^2(3x^2 + e^x)$$

$$\textcircled{2} \quad y = x^{3/2}(x^2 - 1)$$

$$y' = \frac{3}{2} x^{1/2}(x^2 - 1) + x^{3/2}(2x)$$

Quotient rule: p186.

$$\begin{aligned} 1. \quad \frac{d}{dx} \frac{x^2 + \sqrt{x}}{x} &= \frac{d}{dx} \left(\frac{x^2}{x} + \frac{\sqrt{x}}{x} \right) \\ &= \frac{d}{dx} \left(x + x^{-1/2} \right) \\ &= 1 - \frac{1}{2} x^{-3/2} \end{aligned}$$

2. $\frac{d}{dx} \frac{e^x}{x^2} = ?$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

No proof

① $y = \frac{x-3}{x^2}$

$$y' = \frac{1(x^2) - (x-3)2x}{(x^2)^2}$$

② $y = \frac{\sqrt{x}}{x^2 - x}$

$$y' = \frac{\frac{1}{2\sqrt{x}}(x^2 - x) - \sqrt{x}(2x - 1)}{(x^2 - x)^2}$$

③ $y = e^{-x} = \frac{1}{e^x}$

$$y' = \frac{0 - 1 \cdot e^x}{(e^x)^2} = -e^{-x}$$

④ Find the equation of the tangent to $y = \frac{2x}{x+1}$ in the point (1,1)

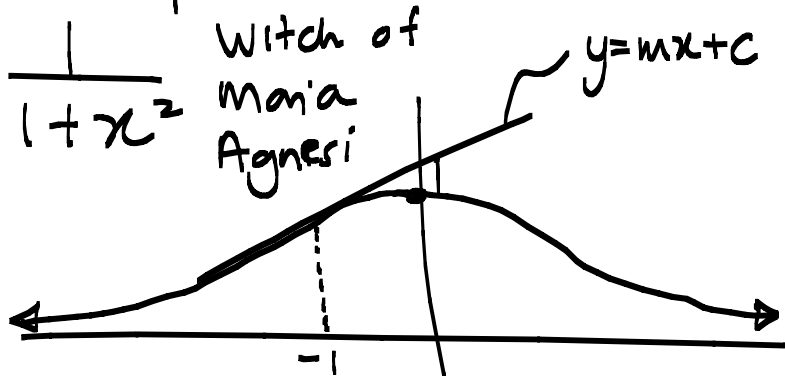
$$y' = \frac{2(x+1) - 2x(1)}{(x+1)^2} = \frac{2}{(x+1)^2}$$

$$y'(1) = \frac{2}{4} = \frac{1}{2} \quad \text{— slope } m = \frac{1}{2} \text{ through}$$

$$(1, 1) \quad y - 1 = \frac{1}{2}(x - 1)$$

Sketch $\frac{1}{1+x^2}$ and find the equation of the tangent line to this curve at the point where $x = -1$.

$$y = \frac{1}{1+x^2} \quad \text{Witch of Maria Agnesi}$$



$$x=0 \Rightarrow y=1$$

$$\lim_{x \rightarrow \infty} \frac{1}{1+x^2} = 0$$

$$\lim_{x \rightarrow -\infty} \frac{1}{1+x^2} = 0$$

$$y' = \frac{-2x}{(1+x^2)^2}$$

$$y'(-1) = \frac{1}{2}$$

through $(-1, \frac{1}{2})$

$$y = \frac{1}{2}x + 1$$